



HEIDELBERG
UNIVERSITY
HOSPITAL



HEIDELBERG
INSTITUTE OF
GLOBAL HEALTH

How LLMs may support selected research tasks – the example of qualitative research

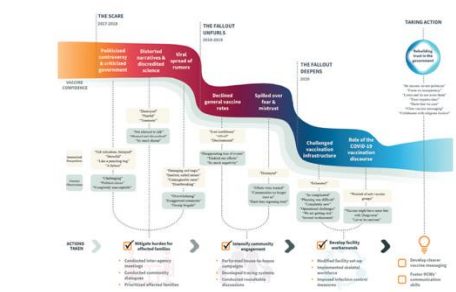
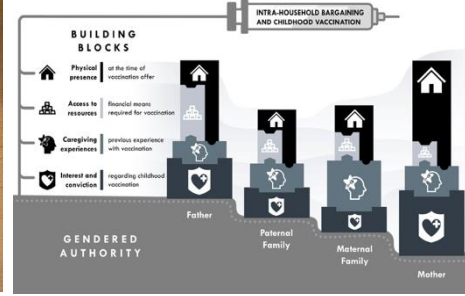
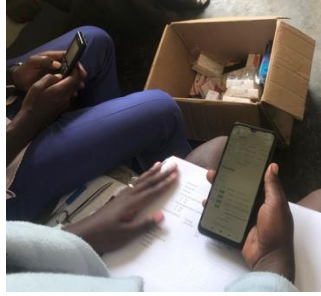
Jonas Wachinger

(with credit to Kate Bärnighausen, Alexandre Infanti, and the DIRGH team)

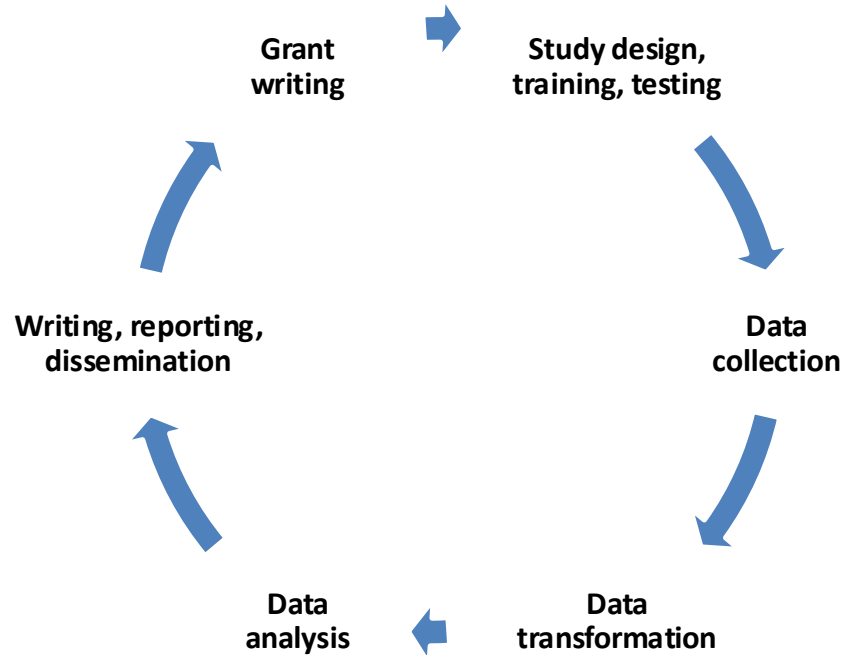
Data Science for Health systems seminar

My perspective on qualitative research:

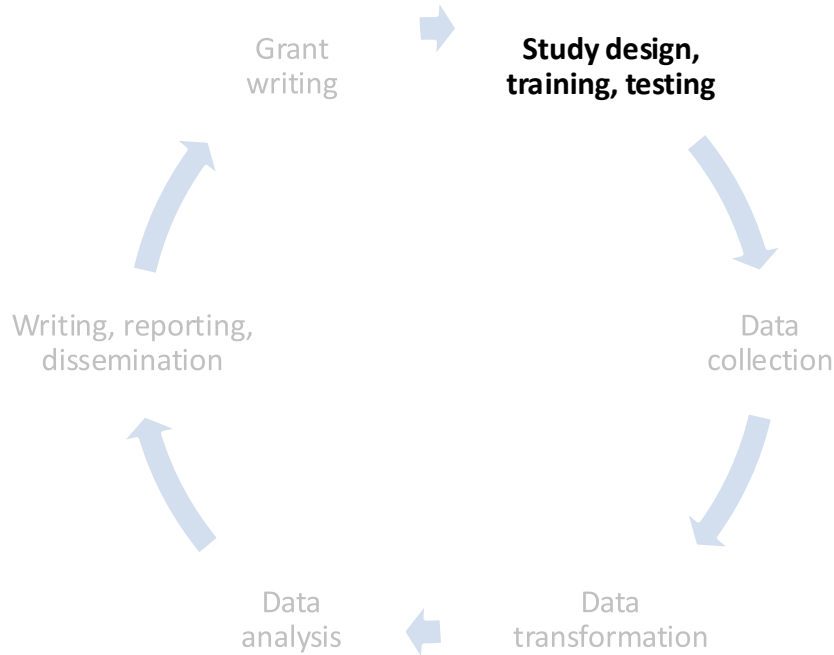
Design and implementation research for global health interventions



LLMs across the qualitative workflow



LLMs across the qualitative workflow



- Draft & refine tools
- Serve as sparring partner
- Support training, create instructional material
- Ideate data collection approaches
- Fake interviews, piloting

LLMs across the qualitative workflow



Out of One, Many: Using Language Models to Simulate Human Samples

Published online by Cambridge University Press: 21 February 2023

Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting and David Wingate

Show author details

Article Figures Supplementary materials Metrics

Save PDF Share Cite Rights & Permissions

Article contents

Abstract
Introduction
The GPT-3 Language Model
Algorithmic Fidelity
Silicon Sampling: Correcting Skewed Marginals
Study 1: Free-Form Partisan Text
Study 2: Vote Prediction
Study 3: Closed-

Abstract

We propose and explore the possibility that language models can be studied as effective proxies for specific human subpopulations in social science research. Practical and research applications of artificial intelligence tools have sometimes been limited by problematic biases (such as racism or sexism), which are often treated as uniform properties of the models. We show that the “algorithmic bias” within one such tool—the GPT-3 language model—is instead both fine-grained and demographically correlated, meaning that proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups. We term this property *algorithmic fidelity* and explore its extent in GPT-3. We create “silicon samples” by conditioning the model on thousands of sociodemographic backstories from real human participants in multiple large surveys conducted in the United States. We then compare the silicon and human samples to demonstrate that the information contained in GPT-3 goes

Access

275
Cited by

Related content

Article

Synthetic Replacements for Human Survey Data? The Perils of Large Language Models
James Baber, Joshua D. Clinton, Casey Dorff, Brenon Konrad, and Jennifer M. Larson

Political Analysis
Published online: 17 May 2024

Article

Panel Effects in the American National Election Studies
Larry M. Bartels
Political Analysis
Published online: 4 January 2017

Article

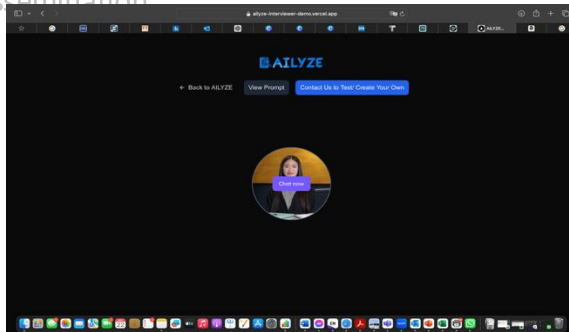
Macropartisanship: An Empirical Reassessment
Paul R. Abramson and Charles W. Ostrom
American Political Science Review
Published online: 1 August 2014

Element

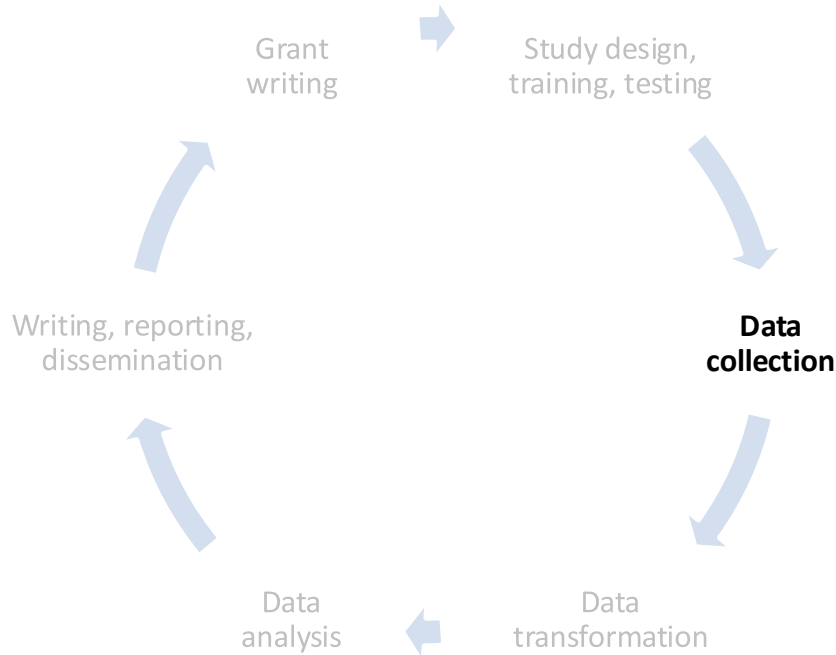
The screenshot shows the QualiaInterviews website. The header includes navigation links: Interviews, Documentation, Blog, Causal Map, Pricing, About Us, and a 'Log in to the app' button. The main content area is titled 'Demo interviews' and includes a sub-header 'How does it actually feel to be interviewed by Qualia?'. Below this, it states 'Below you'll find examples of simple chat interviews with Qualia, which you can click on and try for yourself. No need to login or register.' There are two main sections for 'Try simple chat interviews':

- What went wrong with the project?**
Evaluating a (hypothetical) climate adaptation project and what went wrong with it.
Button: Test the interview yourself
- USA Problems**
An interview about current problems in the USA. A demonstration of 'causal back-chaining'; designed specifically to find reasons for problems, and reasons for the reasons.
Button: Try QualiaInterviews now

On the left side of the interface, there is a vertical list of interview topics with plus signs next to them: '3. Creating an interview', '4. Inviting respondents', and '5. Monitoring and Analysing'.

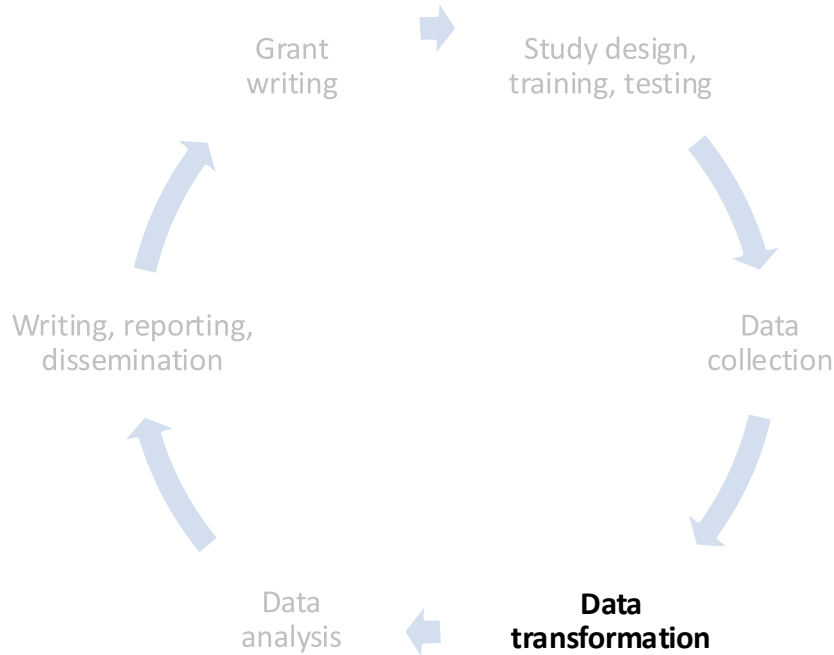


LLMs across the qualitative workflow



- Summarize debriefings
- Serve as debriefing partner
- Condense multimodal data
- Rapidly iterate prototypes

LLMs across the qualitative workflow



- Transcribe data
- Translate data
- Summarize large datasets
- Anonymize qualitative data

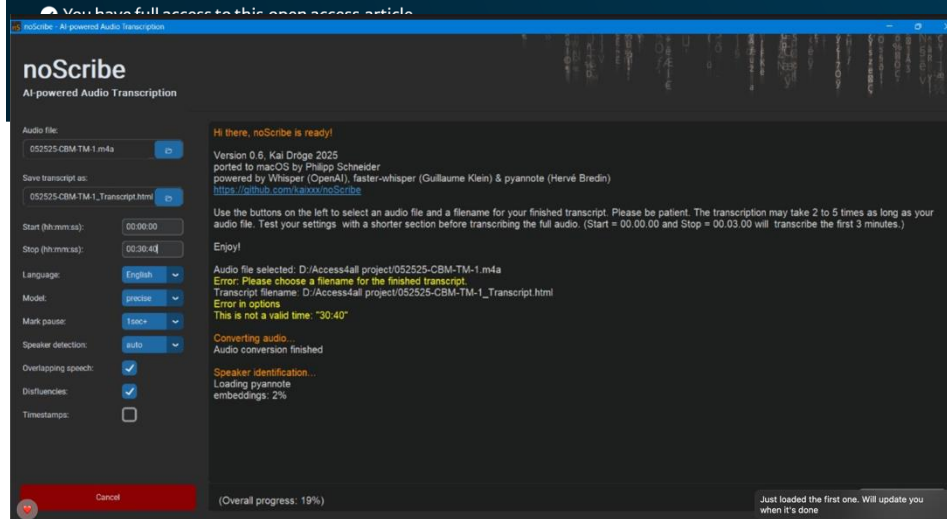
LLMs across the qualitative workflow

Home > BMC Research Notes > Article

From voice to ink (Vink): development and assessment of an automated, free-of-charge transcription tool

Research Note | [Open access](#) | Published: 29 March 2024

Volume 17, article number 95, (2024) | [Cite this article](#)



Data transformation in itself an **interpretive process**

- Not neutral, but involves decisions about what to include, etc.

Embodied and reflexive engagement

- Fosters familiarity and insight

Not a preparatory step, but the **first step of data analysis**

LLMs across the qualitative workflow



Pearls, Pith and Provocation

Prompts, Pearls, Imperfections: Comparing ChatGPT and a Human Researcher in Qualitative Data Analysis

Writing,
dissertation

Qualitative Health Research
2024, Vol. 0(0) 1–16

© The Author(s) 2024



Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/10497323241244669

journals.sagepub.com/home/qhr



Jonas Wachinger¹ , Kate Bärnighausen^{1,2}, Louis N. Schäfer¹ , Kerry Scott³, and
Shannon A. McMahon^{1,3}

analysis



transformation

Identified themes

- Solid performance for descriptive themes
- LLMs can support a (rapid) understanding of qualitative data
- Instances of 'interpretative' themes?

Approach	Qualitative Researcher	ChatGPT analysis				
		General thematic analysis #1	General thematic analysis #2	General thematic analysis #3	Grounded Theory	Framework approach
Themes ^a	• Characteristics and benefits of human patient-doctor interactions ^b	• Patient Perspective				
	• Trust in AI by patients and doctors		• Data Protection and Policy	• Trust and Responsibility	• Data protection concerns • Responsibility of data protection	• Concerns about data protection
	• Diagnostic potential of AI	• Diagnostic Support and Automation	• Potential Benefits of AI	• Potential Benefits of AI	• Perceived benefits of AI	• Potential benefits of AI in healthcare
	• Knowledge requirements	• Challenges and Considerations	• Limited Discussions and Awareness		• Limited awareness and understanding of AI • Need for comprehensive information	
	• Training requirements prior to AI introduction	• Training and Certification	• Training and Certification	• Training and Certification	• Training and testing requirements	• Training and certification
	• General AI perceptions and (medical) use cases	• Perception of AI in Medical Practice	• Perceptions of AI in Medicine	• Perception of AI in Healthcare	• Varied attitudes towards AI among medical professionals	• User perspectives and acceptance
	• Existing use of digital technology and how this might inform AI introduction	• Integration and Implementation				
			• Safety and Reliability	• Acceptance and Resistance	• Certification and supervision	• Responsibility and policy-making
				• General Considerations	• Interest in research findings	

May suggest interpretation & theory engagement...

Box 4. ChatGPT Investigating the Data From a Marxist Perspective.

Applying a Marxist lens to the original analysis of the transcript involves examining the social and economic dynamics at play and analyzing how power, class, and capitalism influence the perspectives and discussions around AI in healthcare. Here's an interpretation from a Marxist perspective:

1. Power dynamics and class struggle:

The mention of trust and the desire for a personal connection between doctors and patients highlights the importance of the doctor–patient relationship. From a Marxist perspective, this relationship can be seen as an expression of power dynamics within the capitalist healthcare system. Doctors hold a position of power and authority, while patients are positioned as recipients of care. The introduction of AI in healthcare raises questions about how power dynamics may shift, potentially reducing the agency and autonomy of patients and reinforcing the dominant position of doctors.

2. Capitalist influence on technology:

The mention of big technology companies and their control over personal health data raises concerns about the capitalist influence on AI in healthcare. Marxist analysis would highlight how capitalist interests drive the development and implementation of AI technologies, with companies seeking to monetize data and maximize profits. This raises ethical considerations around the commodification of health information and the potential exploitation of patients' data for financial gain.

...But does this count as “understanding”?

Box 5. ChatGPT Applying the Vaccine-Specific 5C Model to Its Results.

Certainly! The 5C psychological antecedents of vaccination, as proposed by Betsch and colleagues, are a framework that considers various psychological factors influencing vaccination behavior. Applying this framework to the previous results from the transcript analysis, we can sort and interpret the findings as follows:

1. Complacency:

Limited awareness and understanding of AI: The respondent's limited understanding of AI in healthcare reflects a certain level of complacency, indicating a lack of perceived urgency or need for AI integration in their practice. This suggests that raising awareness and highlighting the potential benefits of AI may be necessary to address complacency.

2. Confidence:

Varied attitudes toward AI among medical professionals: The respondent's mention of colleagues expressing different attitudes toward AI indicates a lack of consensus or confidence in its use. Some colleagues may be more confident and open to adopting AI, while others may express skepticism or resistance. Building confidence in AI technologies through training, evidence-based information, and successful case studies could help address this variation in attitudes.

Risks of hallucinations

Table 2. Themes and Example Quotes.

Quotes provided by ChatGPT	Original text in interview transcript
Perception of AI in Healthcare	
"I think it's definitely a good idea to have AI support for doctors, especially for diagnosis." (R)	- <i>Verbatim quote does not exist in the original transcript; proposed quote maintains fidelity to actual interview content</i>
"People are more and more integrating this [AI] into their lives. It's always a question of data safety... but I think it has its useful sides." (R)	"I think people are more and more integrating this into their lives. It's always a question of data safety if you want technicals, it's not the medical system that gets these data, but big technology companies. (ehem) So I think that's a large question, but I think it has also its useful sides. " (R)
Potential Benefits of AI	
"If those technologies [wearables] have had to diagnose maybe some disease that we wouldn't have found in another way... it saves lives." (R)	"People like who are asking for you to use that, if they have maybe some wearables, some smartwatch or something like that, and if those technology have had to diagnose maybe some disease that we wouldn't have found in another way. And like maybe in the end, how patients save lives. " (R)
"It can definitely improve accuracy, especially in difficult cases or early detection." (R)	- <i>Verbatim quote does not exist in the original transcript; proposed quote maintains fidelity to actual interview content -</i>

A lot has happened since...



Why MAXQDA? ▾

Products ▾

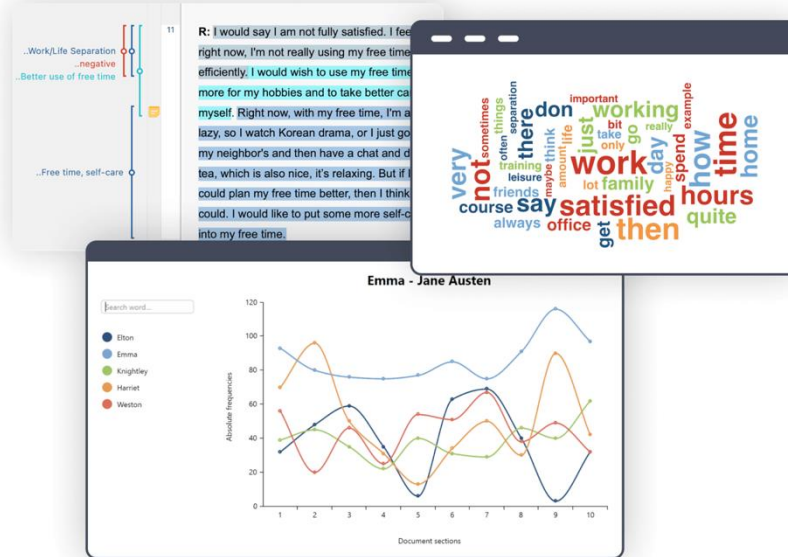
Resources ▾

Pricing

Support ▾



Free Trial



A lot has happened since...

7

Generative AI: A Catalyst for Paradigmatic Change in Qualitative Data Analysis

Author: [Susanne Friese](#) | [AUTHORS INFO & AFFILIATIONS](#)

Publication: Artificial Intelligence (AI) in Social Research • pp. 71-83 • <https://doi.org/10.1079/9781800626607.0007>

28 1 0 0 0

GET ACCESS

Abstract

The aim of this chapter is to investigate whether Thomas Kuhn's concept of scientific revolutions, as described in his seminal work *The Structure of Scientific Revolutions* (1962), can be extended to examine shifts in qualitative research methodologies driven by the advent of artificial intelligence (AI). Initially, Kuhn's process of scientific revolution will be outlined, focusing primarily on the dynamics detailed in his landmark text. While Kuhn's discussion was predominantly confined to the natural sciences, there is ongoing debate about its applicability to the social sciences. This debate will be addressed, arguing for the framework's relevance beyond the natural sciences, as Kuhn himself suggested in a postscript to a later edition (2012). Building on this foundation, the history of one of the predominant methodologies in qualitative data analysis, coding, will be explored. The exploration will consider how the emergence of AI technology could potentially induce a Kuhnian-type shift, leading to significant transformations in how qualitative data will be analysed in the future.



Help



A lot has happened since...

Original Article

We Reject the Use of Generative Artificial Intelligence for Reflexive Qualitative Research

Tanisha Jowsey¹, Virginia Braun², Victoria Clarke³,
Deborah Lupton⁴, and Michelle Fine⁵

Abstract

Four hundred and nineteen experienced qualitative researchers from 32 countries invite readers of *Qualitative Inquiry* to consider their position on use of generative artificial intelligence (GenAI) for qualitative research. We hold the position that analytic approaches such as reflexive thematic analysis are human research practices requiring a subjective, positioned, and reflexive researcher and therefore the use of GenAI in such approaches is not methodologically congruent. We additionally reject GenAI for reflexive qualitative approaches on the grounds of social and environmental justice.

Qualitative Inquiry

1–5

© The Author(s) 2025



Article reuse guidelines:

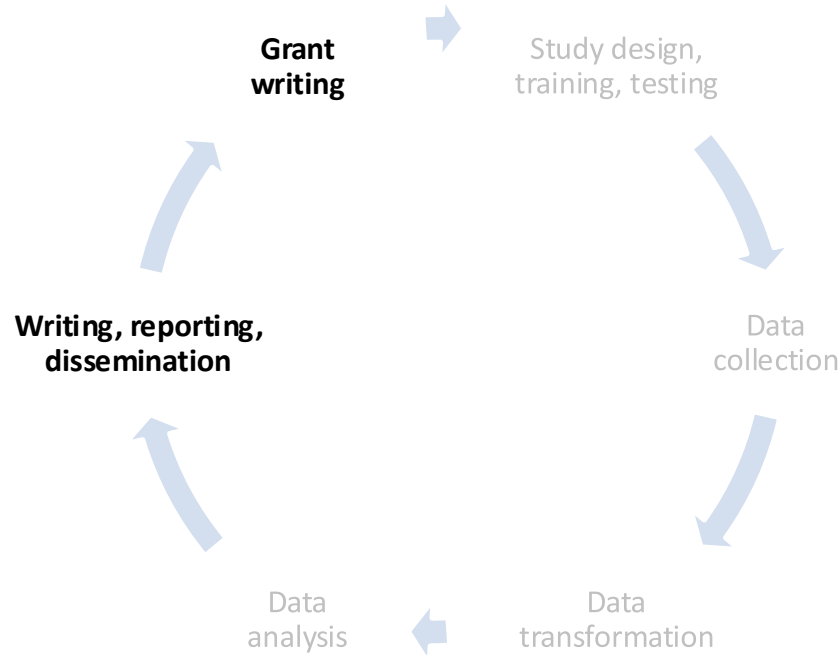
sagepub.com/journals-permissions

DOI: 10.1177/10778004251401851

journals.sagepub.com/home/qix



LLMs across the qualitative workflow



- Identify research problems
- Challenge ideas
- Estimate budgets
- Generate text
- Provide writing guidance
- Draft visualizations

Our (pragmatic) thinking – as of now

- We cannot be sure whether (and how) LLMs may “understand meaning”
- Ethical challenges remain (bias, confidentiality, accessibility, resource use...)
- LLMs won't go away, and people will use them
- Expecting LLMs to exactly “apply” existing analytic approaches and paradigms likely won't work
- If we stay mindful, there may be several use-cases for LLMs as tools across the qualitative workflow

If you want to read more...

Original Research Article

Integrating Artificial Intelligence Within the Qualitative Workflow: An Introductory Primer

International Journal of Qualitative Methods

Volume 25: 1–20

© The Author(s) 2026

DOI: 10.1177/16094069261495187

journals.sagepub.com/home/ijq



Alexandre Infanti^{1,2} , Jonas Wachinger¹ , Kate Bärnighausen¹ , Mark Donald C. Reñosa^{1,3}, and Shannon A. McMahon¹ 

Thanks!

Jonas.wachinger@uni-heidelberg.de



DIRGH team



Publication

With thanks to: Kate Bärnighausen, Alexandre Infanti, Shannon McMahon, Kerry Scott, Mark Reñosa, Louis Schäfer, and the broader DIRGH and HIGH communities